



VERITAS UNIVERSITY, ABUJA

DEPARTMENT OF SOFTWARE ENGINEERING
VUA-SEN 201: SOFTWARE REQUIREMENTS ENGINEERING
2025/2026 SESSION Theory

Time: 2hrs

Answer **QUESTION 1** and any other **TWO** questions of your choice from the following questions. All questions carry equal marks.

Question 1 ✓

- a. Write in detail what you understand by Software requirement engineering and its importance to software engineers. (10 Mks).
- b. A university is developing an e-campus system that covers online registrations, hostel allocations, online payments library management system, and examination systems.
 - i. Identify at least **five stakeholder groups**. (5mks)
 - ii. Choose three elicitation techniques for each group and justify your choice (5mks)
 - iii. Explain one problem that may arise from not involving all stakeholders. (5mks)
 - iv. Develop a simple requirement specification for one of the stakeholders showing three functional and two non functional. (5mks)
- c. While developing a student results processing system, the university management suddenly requests an additional feature to include **automated GPA alerts**.
 - i. What type of Requirements Engineering issue is this? (2Mks)
 - ii. Why should such changes be properly managed? (2mks)
 - iii. Mention one possible effect of uncontrolled requirement changes. (2mks)
 - iv. Differentiate between requirements elicitation and requirements management. (4mks)

Questions: 2 ✓

A private hospital intends to develop a *web-based patient management system* to handle appointments, billing, electronic medical records, and reporting. Management is concerned about cost, technical capability, and system sustainability.

- a) Explain the purpose of conducting a feasibility study before system development. (6 marks)
- b) Analyze the following feasibility aspects for the proposed system:
 - i. Technical feasibility
 - ii. Economic feasibility
 - iii. Operational feasibility (12 marks)
- c) Explain how cost-benefit analysis can guide management decision-making. (6 marks)
- d) Identify two risks that may arise if feasibility analysis is poorly conducted. (6 marks)

Question 3:

A **fintech startup** is developing a **Mobile Loan Application System** that allows users to apply for loans, upload documents, receive credit scoring results, and track repayment schedules. During development, stakeholders complain that:

- Some implemented features do not meet business expectations.
- Performance is slow during peak usage.
- Regulatory compliance requirements were not fully captured.
- Some requirements were ambiguous.

As the Software Engineer, answer the following:

- a) Identify and explain four validation problems evident in this scenario. (8 marks)
- b) Describe and explain four requirements validation techniques that could have prevented these issues. (12 marks)
- c) Explain the difference between requirements verification and validation, and apply both concepts to this system. (6 marks)
- d) Discuss the consequences of poor requirements validation on software quality and project cost. (4 marks)

Question 4 ✓

- (a) Given the requirement: *"The system shall provide real-time traffic updates."*

As requirement engineer, Break it into at least **four refined, testable requirements**. (10 marks)

- (b) Rewrite the following poorly written requirement: *"The system should be secure and easy to use."* (10 mks)
- (c) Write short notes on Requirement Analysis and Nonfunctional requirements (10 marks)

Question 5

- a) Define *Software Engineering* and explain its core objectives. (3 marks)
- b) Describe the **Waterfall** and **Agile** software process models. (6 marks)
- c) Compare Waterfall and Agile models, stating two advantages and two disadvantages of each. (6marks)
- d) Explain how the choice of a software process model affects **risk management** in software projects. (6marks)
- e) A hospital system passed technical tests but was rejected by nurses due to poor workflow support. Early validation could have prevented this failure. Identify what should have been done and propose solution. (9mks)